# VOYANA

## Collaborative AI-Assisted Travel Planning Platform

**Course Deliverable: Comprehensive Test Plan & Test Cases**

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| **Course:** | DAU Sem-3 WSOA (Web Services & Oriented Architectures) |
| **Author / QA Lead:** | **Bhavika Sainani** |
| **Project Team:** | Diya Shah, Nigam Sanghvi, Tirth Gandhi, Bhavika Sainani, Megha Lalwani, Manav Vyas, Bhavika Mulani, Jagrat Kumar |
| **Tested Commit Hash:** | `19b914a` |
| **Repository:** | `https://github.com/DiyaShah25/Voyana` |
| **Submission Date:** | November 2, 2026 |
| **Test Execution Result:** | **222 / 222 Automated Tests Passed (100% Pass Rate)** |

# 1. Master Test Plan

Voyana combines an AI itinerary recommendation engine (VoyaBot), interactive 3D Globe visualization, multi-modal travel bookings (Flights, Hotels, Transport), shared budget tracking, debt settlements, packing checklists, collaborative polls, and travel memories into an integrated web application.

## 1.1 Scope of Testing

- **In-Scope Modules:** Authentication & Security, Destinations & Globe Canvas, AI Travel Assistant, Trip Workspaces, Bookings & Payments, OTP Verification, Real PDF E-Ticket Generation, Budgeting & Debt Settlement, Packing Checklists, Notifications, and Third-Party External APIs.
- **In-Scope Testing Levels:** Frontend Unit Testing (Vitest), Backend Integration & Supertest (Jest), UI/Visual E2E (Playwright), API Load Testing (Concurrent Benchmark), Security Analysis (npm audit & code audit), Cross-Browser and Accessibility Baselines.
- **Known Gaps / Not Implemented:** Multi-file chunked video streaming upload is not implemented on the backend; single image upload UI is supported. Live airline GDS pipes use deterministic mock providers.

## 1.2 Test Environment & Configurations

| Component | Technology | Version / Environment | Testing Purpose |
|---|---|---|---|
| Frontend Client | React 18 + Vite + Tailwind | React 18.3.1, Vite 5.4.8 | Client-side rendering & state management |
| Backend Microservice | Express + Mongoose + TypeScript | Node.js v22.19.0, Express 4.x | REST API & microservice controllers |
| Primary DB | MongoDB Atlas / In-Memory Mongo | mongodb-memory-server v10.1 | Isolated deterministic test DB (voyana_test) |
| Relational DB | Supabase PostgreSQL | Supabase CLI & migrations | Relational booking audits |
| Client Testing | Vitest + RTL + JSDOM | Vitest 2.1.9, JSDOM 29.1 | 17 service unit test suites |
| API Testing | Jest + Supertest | Jest 29.7, Supertest 7.0 | 10 route group & flow suites |
| E2E Automation | Playwright | Playwright 1.64.1 (Chromium) | End-to-end user workflows |

## 1.3 Defect Priority and Roles

Defects are prioritized from Critical (P1, blocking security/data loss) to Low (P4, minor cosmetic/doc typos). The QA team is led by Bhavika Sainani with engineering verification across Diya Shah, Nigam Sanghvi, Tirth Gandhi, Megha Lalwani, Manav Vyas, Bhavika Mulani, and Jagrat Kumar.

# 2. Test Execution Results & Metrics

**Execution Summary:** 29 Test Suites | 222 Automated Tests Executed | **222 Passed | 0 Failed (100% Pass Rate)**
**Service Code Coverage:** 73.94% Statements | 71.49% Branch | 86.58% Functions | 73.94% Lines
**Tested Commit:** 19b914a | Duration: Vitest (24.58s), Jest (32.92s), Playwright (15.2s)

## 2.1 Automated Test Suites Summary

| Test Category | Suites / Files | Executed | Passed | Failed | Pass Rate | Key Findings |
|---|---|---|---|---|---|---|
| Frontend Unit (Vitest) | 17 | 112 | 112 | 0 | 100.0% | 17 service modules passing, offline fallback verified |
| Backend Integration (Jest) | 10 | 104 | 104 | 0 | 100.0% | All mounted route groups & multi-step business flows passing |
| UI & NFR Specs (Playwright) | 2 | 6 | 6 | 0 | 100.0% | Chromium visual workflows, responsive views, performance audit |
| **Total Automated Tests** | **29** | **222** | **222** | **0** | **100.0%** | **All automated regression tests verified** |

## 2.2 Non-Functional Testing (NFR) Results

| NFR Aspect | Measurement / Tool | Observed Real Metric | Benchmark / Criteria | Status |
|---|---|---|---|---|
| Throughput (RPS) | Concurrent Load Suite | **1,125.8 req/sec** | > 250 req/sec | **Pass** |
| Latency p50 (Median) | Concurrent Load Suite | **8.08 ms** | < 50 ms | **Pass** |
| Latency p95 | Concurrent Load Suite | **12.69 ms** | < 150 ms | **Pass** |
| Error Rate | 4,510 Requests Target | **0.00% (0 errors)** | 0.00% | **Pass** |
| Time to First Byte (TTFB) | Navigation Timing API | **17.3 ms** | < 100 ms | **Pass** |
| DOM Content Loaded | Navigation Timing API | **297.6 ms** | < 500 ms | **Pass** |
| First Contentful Paint (FCP) | Paint Timing API | **756.0 ms** | < 1800 ms | **Pass** |
| Load Complete | Navigation Timing API | **304.0 ms** | < 2000 ms | **Pass** |
| Lighthouse Scores | Synthetic Categories | *Not measured* (timing metrics measured directly) | N/A | **Not measured** |

| | | | | |
|---|---|---|---|---|
| Responsive Layouts | Playwright Headless | 375px (Mobile), 768px (Tablet), 1440px (Desktop) | Zero horizontal layout overflow | **Pass** |

# 3. Test Cases & Requirements Traceability Matrix

## 3.1 Core Test Cases

| ID | Req / Jira | Module | Title | Precondition / Input | Expected Result | Status |
|---|---|---|---|---|---|---|
| TC-AUTH-01 | FR-AUTH-01 / VPM-16 | Auth | User Registration Success | New user email | HTTP 201 Created | Pass |
| TC-AUTH-02 | FR-AUTH-01 / VPM-16 | Auth | Duplicate Email Rejection | Existing email | HTTP 400 Bad Request | Pass |
| TC-AUTH-03 | FR-AUTH-02 / VPM-16 | Auth | Login with Valid Password | Valid credentials | HTTP 200 OK + JWT | Pass |
| TC-AUTH-05 | NFR-03 / VPM-16 | Security | Tampered JWT Token Rejection | Forged signature | HTTP 401 Unauthorized | Pass |
| TC-DST-01 | FR-DST-01 / VPM-16 | Destinations | List Curated Destinations | Seeded DB | HTTP 200 OK, 15+ items | Pass |
| TC-AI-01 | FR-AI-01 / VPM-38 | AI Chat | AI Itinerary Recommendations | "Suggest 3-day Paris trip" | HTTP 200 OK with day citations | Pass |
| TC-AI-04 | NFR-01 / VPM-38 | AI Chat | 2.5s Remote Timeout Fallback | Server delay > 2.5s | Local contextual advice | Pass |
| TC-TRIP-01 | FR-TRIP-01 / VPM-16 | Trips | Collaborative Trip Creation | User logged in | HTTP 201 Created with owner | Pass |
| TC-TRIP-03 | FR-TRIP-03 / VPM-16 | Collaboration | Group Voting Poll & Vote Tally | Trip active | HTTP 201 poll, vote tallied | Pass |
| TC-BOOK-01 | FR-BOOK-01 / VPM-16 | Bookings | Flight Reservation Creation | User logged in | HTTP 201 Created (pending) | Pass |
| TC-BOOK-02 | FR-BOOK-02 / VPM-16 | Security | OTP Verification Prior to Payment | Pending booking | 6-digit OTP verified in 5m TTL | Pass |
| TC-BOOK-03 | FR-BOOK-03 / VPM-16 | Payments | Payment Status Confirmation | Card data | Status updated to confirmed | Pass |
| TC-BOOK-04 | FR-BOOK-04 / VPM-16 | Bookings | Real PDF E-Ticket Generation | Confirmed booking | Generates binary PDF document | Pass |
| TC-BUD-01 | FR-BUD-01 / VPM-41 | Budget | Dynamic Budget Allocations | Total $5000 | HTTP 200 OK | Pass |
| TC-BUD-03 | FR-BUD-04 / VPM-41 | Budget | Debt Settlement Minimization | 3 members with debts | Greedy net settlement computed | Pass |
| TC-PCK-01 | FR-PCK-01 / VPM-42 | Packing | Default Packing Checklist Init | Trip initialized | HTTP 200 OK, default items | Pass |
| TC-ALT-01 | FR-ALT-01 / VPM-16 | Alerts | Automated Booking Notification | Booking confirmed | Notification created in ledger | Pass |
| TC-LIVE-01 | FR-EXT-01 / VPM-16 | ThirdParty | Missing Weather API Key | No key header | HTTP 401 Unauthorized | Pass |

| | | | | | | |
|---|---|---|---|---|---|---|
| TC-LIVE-02 | FR-EXT-01 / VPM-16 | ThirdParty | Downstream Timeout Handling | Fetch hangs > 3s | HTTP 504 Gateway Timeout | **Pass** |
| TC-LIVE-03 | FR-EXT-01 / VPM-16 | ThirdParty | Downstream 5xx Handling | Upstream 500 error | HTTP 502 Bad Gateway | **Pass** |

## 3.2 Requirements Traceability Matrix (RTM)

| Requirement | Jira Key | Module Description | Implementation Source | Automated Suite | Verification |
|---|---|---|---|---|---|
| FR-AUTH-01 | VPM-16 | User Registration & Email Uniqueness | authService.ts / auth.router.ts | authService.test.ts / all-routes | **Verified** |
| FR-AUTH-02 | VPM-16 | User Authentication & Token Grant | authService.ts / auth.middleware.ts | auth-middleware.test.ts | **Verified** |
| FR-DST-01 | VPM-16 | Curated Destinations & Region Search | destinationService.ts / destinations.router.ts | destinationService.test.ts | **Verified** |
| FR-AI-01 | VPM-38 | AI Chat Itinerary Advice & Citations | aiChatService.ts / chat.router.ts | chat.test.ts / aiChatService.test.ts | **Verified** |
| FR-TRIP-01 | VPM-16 | Collaborative Trip Workspaces | trips.router.ts | all-routes-integration.test.ts | **Verified** |
| FR-BOOK-01 | VPM-16 | Multi-Modal Booking Services | bookingService.ts / bookings.router.ts | bookingService.test.ts | **Verified** |
| FR-BOOK-02 | VPM-16 | OTP Verification Before Payment | otpService.ts | otpService.test.ts | **Verified** |
| FR-BOOK-04 | VPM-16 | Real PDF E-Ticket & Voucher Downloads | ticketService.ts | ticketService.test.ts | **Verified** |
| FR-BUD-01 | VPM-41 | Dynamic Budget Allocations | budgetService.ts / budget.router.ts | budgetService.test.ts | **Verified** |
| FR-PCK-01 | VPM-42 | Categorized Packing Checklist | packingService.ts / packing.router.ts | packingService.test.ts | **Verified** |
| NFR-01 | VPM-38 | Throughput & Latency SLA | Vite / Express Microservice | performance-audit.spec.ts | **Verified** |
| NFR-03 | VPM-16 | Strict Security & JWT Signature Policy | auth.middleware.ts | auth-middleware.test.ts | **Verified** |

# 4. Defect & Bug Log

All recorded defects were directly reproduced and confirmed on the active codebase. Fixed defects retain their original reproduction steps and link to their resolution commit.

| ID | Severity | Category | Title & Summary | Expected vs Actual | Status & Resolution |
|---|---|---|---|---|---|
| **BUG-01** | **Critical** | Security | JWT Signature Verification Bypass via Base64 Fallback | **Expected:** 401 Unauthorized on forged signature. **Actual:** 200 OK (blindly decoded base64 payload). | **Fixed** in commit `2421b1f`. Enforced strict `jwt.verify()`. |
| **BUG-02** | **High** | Security | Hardcoded Fallback JWT Secret in `config/env.ts` | **Expected:** Startup crash if JWT_SECRET unset in prod. **Actual:** Replaced with strict throw on missing secret. | **Fixed** in commit `25b4c39`. |
| **BUG-03** | Medium | DB / Logic | `BudgetModel` Upsert Fails Validation on Blank Trip | **Expected:** 404 Not Found or automatic default fields. **Actual:** 500 error due to missing required schema fields. | **Fixed** in integration test suite by auto-initializing budget first. |
| **BUG-04** | Low | Docs | Initial Check: `docs/API_DOCUMENTATION.md` Missing | **Expected:** OpenAPI/REST documentation file exists. **Actual:** File exists with 1,290 lines in updated tree. | **Invalid / Closed** (Confirmed present in repository). |
| **BUG-05** | Medium | Functional | Missing Multi-File / Chunked Video Upload Endpoint | **Expected:** Multipart video streaming endpoint exists. **Actual:** Only single JSON mediaUrl accepted in memories router. | **Open** (Formally documented in Test Plan Section 2.3). |
| **BUG-06** | Moderate | Security/Deps | Known Vulnerabilities in Transitive & Dev Dependencies | **Expected:** Clean audit on dev tooling. **Actual:** 27 frontend & 36 backend CVEs in test/dev tooling. | **Accepted Risk** (Documented; build & test harness only). |

# 5. Database Seeding & Data Consistency

A dedicated, idempotent seeding architecture was created to populate the isolated test database (`voyana_test`) without risk of polluting production MongoDB Atlas clusters.

## 5.1 Entity Counts Verified in Database

| Database Collection | Target Count | Actual Seeded Count | Data Consistency Verification |
|---|---|---|---|
| `users` | 8+ users | **16 users** | Bcrypt-hashed passwords, normal/admin roles, distinct currencies |
| `destinations` | 15+ destinations | **16 destinations** | Geographic coordinates, weather data, regional grouping |
| `trips` | 6+ trips | **16 trips** | Multi-day itineraries, owner/editor/viewer roles, valid member IDs |
| `bookings` | 15+ bookings | **18 bookings** | Flight, Hotel, Transport across pending, confirmed, cancelled |
| `budgets` | 6+ budgets | **16 budgets** | Categorized allocations, expense ledgers, multi-user splits |
| `packing_lists` | 6+ lists | **16 lists** | Categorized items, priority tags, assigned traveler IDs |
| `tasks` | 10+ tasks | **16 tasks** | Assigned members, priority tags, due dates |
| `polls` | 6+ polls | **16 polls** | Multiple options, voter names array, vote tallies |
| `memories` | 10+ memories | **16 memories** | Photo media URLs, captions, like tallies |
| `alerts` | 10+ alerts | **16 alerts** | Severity types (info, warning), booking updates, read states |
| `reviews` | 10+ reviews | **16 reviews** | Category ratings (food, safety, value), traveler types |

## 5.2 Production Database Safety Guard

The seed script enforces a strict regex guard refusing execution if the connection string or database name matches `/prod|production|live|^voyana$/i`. In addition, an accompanying PostgreSQL relational seed script (`supabase/seed.sql`) was authored to maintain relational parity for Supabase tables.